Anubhav Chakraborty, PhD

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Professional Summary

Molecular biologist and genetic engineer with a PhD in Molecular and Integrative Physiology (awarded August 2026), currently a postdoctoral fellow at the University of Kansas Medical Center working toward gene therapy for Autosomal Dominant Polycystic Kidney Disease. Experienced with gene perturbation and genome engineering, epigenetic editing and modifications, development of cellular models, gene delivery through viral and non-viral modalities, and design of quantitative assays across primary cells and other hard-to-transduce mammalian cell types. Can independently design, run, analyze, and interpret experiments. Learns new methods quickly and is drawn to technologies that make cell studies easier and more reproducible. Thinks critically, stays steady under pressure, and works well in fast-paced, collaborative research environments where several projects and priorities run at once. Productive scientist with peer-reviewed publications, a U.S. provisional patent, competitive fellowship funding, national presentations, and a record of mentoring junior researchers and collaborating across labs. Aims to grow into a better scientist and thinker who can lead discoveries toward faster and better therapies, ways to prevent disease, or ways to change cell fate.

Education/Employment

University of Kansas Medical Center — **Kansas City, KS, USA**
Postdoctoral Fellow, Jared Grantham Kidney Institute — *Oct 2026 – Present*

University of Kansas Medical Center — **Kansas City, KS, USA**
Doctor of Philosophy (PhD) in Molecular and Integrative Physiology — *Aug 2019 – Aug 2026*
Cumulative GPA: 3.88/4.0

Vellore Institute of Technology — **Tamil Nadu, India**
Council of Scientific and Industrial Research (CSIR) - Govt. of India, — *Dec 2017 – Jun 2019*
Senior Research Fellow

SRM Institute of Science and Technology — **Tamil Nadu, India**
Master of Technology (MTech) in Genetic Engineering — *Jun 2015 – May 2017*
First Class (CGPA: 7.23/10)

SASTRA University — **Tamil Nadu, India**
Bachelor of Technology (BTech) in Bioengineering — *Jun 2011 – May 2015*
First Class (CGPA: 7.33/10)

Technical Skills & Scientific Capabilities

- **Genome Perturbation & Gene Regulation:** Prime editing, CRISPR-Cas9/Cas12a editing, Epigenetic editing and modulation, CRISPR activation (CRISPRa) and CRISPR interference (CRISPRi), guide RNA and pegRNA design, reporter-based and arrayed guide RNA screening, ATAC-qPCR, genomic DNA extraction and edit validation, chromatin accessibility assays.
- **Molecular Cloning & Vector Engineering:** Restriction-ligation cloning, Gibson assembly, Golden Gate / Type IIS cloning, plasmid engineering, mini-circle production, inducible expression systems (IPTG, Tet-On, Tet-Off), protein and mRNA trans-splicing systems, CRISPR ribonucleoprotein (RNP) synthesis, construct QC and sequence verification (Sanger, Nanopore).
- **Gene Delivery:** rAAV, lentivirus, and MMLV design and production, DNA transposon, virus-like particles (VLPs), nucleofection of RNP, mRNA and plasmid (including arrayed formats), lipid-based transfection reagents, PEI transfection.
- **RNA Biology:** In vitro transcription of sgRNAs, column-based RNA purification, RT-qPCR-based RNA stability assays, synthetic mRNA transfection and protein expression analysis.

- **Cell Models & Culture:** Primary human and mouse renal epithelial cell isolation and culture, patient-derived primary cells, cryopreservation, 3D Matrigel culture, hanging-drop spheroids, co-culture and feeder-layer systems, stable and monoclonal cell line generation.
- **Cell-Based Assays & Phenotyping:** Dose-response and EC50 analysis, viability and cytotoxicity assays (CellTiter-Glo, LDH-Glo, MTT, CCK-8, Alamar Blue), apoptosis and necroptosis assays (Annexin V, Caspase-3/7, TUNEL), mitochondrial membrane potential (MitoTracker dyes), ER stress readouts, fluorescent and bioluminescent reporter assays, flow cytometry (GFP/RFP expression on Attune NxT), 3D cyst-growth assays.
- **Molecular & Protein Analysis:** Western blotting, ELISA, immunofluorescence and immunocytochemistry, RT-qPCR, PCR, gel electrophoresis, protein maturation and glycosylation analysis (Endo H, PNGase F).
- **Animal Models:** *Hands-on experience from 2017–2019; can refresh these skills quickly.* Mouse renal ischemia-reperfusion injury surgery, anesthesia, tissue collection and necropsy, tissue microdissection, intraperitoneal injection.
- **Software & Digital Tools:** Benchling, SnapGene, GraphPad Prism, FlowJo, FIJI (microcyst image analysis), Image-Pro Premier, Ingenuity Pathway Analysis (IPA), LabArchive (ELN), Microsoft Office.

Research Experiences

The Jared Grantham Kidney Institute, University of Kansas Medical Center **Kansas City, KS, USA**
Postdoctoral Fellow | Lab of Dr. Alan S. Yu *Oct 2026 – Present*

- **Expanding delivery strategies:** Designing and evaluating rAAV-based delivery of mRNA and protein trans-splicing machinery to support CRISPR-Cas9-based *PKD1* perturbation in primary and immortalized cells and in mouse models of ADPKD.
- **Building stable, validated cell models:** Using a DNA transposon system and prime editing to insert constructs at safe-harbor loci, creating consistent ADPKD cell lines for testing disease mechanisms and interventions.
- **Measuring the cost of delivery:** Designing experiments to assess toxicity from nucleic acid delivery, cellular stress responses to rAAV capsids, and molecular changes after *PKD1* perturbation.

The Jared Grantham Kidney Institute, University of Kansas Medical Center **Kansas City, KS, USA**
PhD Researcher | Lab of Dr. Alan S. Yu *Aug 2020 – Aug 2026*

- **Developed a CRISPRa strategy for ADPKD:** Designed and tested CRISPR activation of endogenous *PKD1*, which raised polycystin-1 levels and altered cellular features of disease. This work led to a proof-of-concept for a gene therapy for ADPKD, a U.S. provisional patent application, and peer-reviewed publications.
- **Tackled epigenetic barriers:** Investigated chromatin accessibility and heterochromatin at the *PKD1* promoter by ATAC-qPCR and benchmarked dCas9 effectors such as p300, Tet1, VPR, SAM, VP64 and endogenous transcription factors. Screened guide RNAs in an arrayed, reporter-based format and identified the optimal sgRNA for CRISPRa of endogenous *PKD1*.
- **Explored pharmacologic approaches:** Tested trichostatin and retinoic acid as alternative pharmacologic and gene-regulation approaches to increase *PKD1*, both on their own and in combination with CRISPRa.
- **Engineered and tested 50+ constructs:** Built and verified reporter and expression constructs through multiple cloning strategies including multi-fragment cloning via Gibson and Golden Gate Assembly, and compared polycystin-1 expression from different mRNA and cDNA designs to see how payload design shapes protein output.
- **Turned a failed method into a better one:** After poor VSV-G lentiviral transduction in mouse renal epithelial cells, used the results to develop a non-viral approach that improved transgene delivery.
- **Solved a delivery bottleneck in vitro:** Developed a one-pot Type IIS restriction/ligation strategy that reduced a CRISPRa plasmid from about 14.9 kb to 6 kb and improved transgene delivery roughly 3-fold in difficult-to-transfect primary renal epithelial cells.
- **Benchmarked delivery platforms:** Evaluated and optimized nucleofection, lipid-based transfection, VLPs, lentivirus, and rAAV for delivering Cas9/Cas12a RNP, sgRNA, mRNA, and plasmid payloads into primary and immortalized human and mouse renal epithelial cells.

- **Prepared RNP components in-house:** Synthesized sgRNAs by *in vitro* transcription for Cas9 RNP synthesis.
- **Built disease models and quantitative assays:** Established primary human and mouse renal epithelial cultures (including patient-derived cells), 3D Matrigel microcyst assays, and hanging-drop spheroids. Developed fluorescent reporters, ELISA, flow cytometry, viability, cytotoxicity, mitochondrial membrane potential, and ER stress readouts.
- **Tested drug responses in cells:** Ran dose-response and EC50 experiments with doxycycline to tune inducible gene expression systems, trichostatin to probe epigenetic changes, and antibiotics to set kill curves and measure cellular toxicity.
- **Characterized protein output:** Analyzed polycystin-1 maturation and glycosylation by Western blotting and Endo H/PNGase F assays.
- **Kept results reproducible in variable cells:** Used matched controls and replicates and validated findings across independent cell lines, donors, and experiments, including in patient-derived primary cells.
- **Contributed to collaborative studies:** Helped investigate inflammatory signaling, extracellular matrix remodeling, and tissue remodeling in early PKD, including bulk RNA-seq analysis.
- **Advised and mentored:** Served as a go-to resource for other labs on bespoke viral vector design, plasmid engineering, and stable cell line derivation using transwell feeder-layer and co-culture systems, and trained 4 junior researchers

Vellore Institute of Technology (VIT) **Tamil Nadu, India**
Senior Research Fellow (CSIR) | Lab of Dr. Pragasam V *Dec 2017 – Jun 2019*

- **Established a mouse model of acute kidney injury:** Built and characterized a renal ischemia-reperfusion injury model, performing surgeries in 50+ mice, including randomization, anesthesia, tissue collection, necropsy, and evaluation of injury.
- **Studied epigenetics in kidney injury:** Investigated DNA methylation and demethylation dynamics linked to necroptotic cell death, using 5mC and 5hmC ELISA along with Annexin V, Caspase-3/7, TUNEL, and DNA fragmentation assays.
- **Supported grants and training:** Helped faculty draft competitive grant proposals and co-directed the thesis work of 4 master's students.

SRM Institute of Science and Technology **Tamil Nadu, India**
Master's Researcher | Lab of Dr. J. Megala, Human Genetics Lab *Jan 2016 – May 2017*

- **Ran a large genotyping study:** Genotyped 1,100 clinical subjects to identify *TNF-alpha* promoter variants associated with *H. pylori*-related gastric mucosal inflammation.
- **Mapped transcriptional regulation:** Characterized phylogenetic footprints and transcriptional profiles of mutated *TNF-alpha* variants linked to chronic inflammation.

SASTRA University **Tamil Nadu, India**
Bachelor's Researcher | Lab of Dr. C. David Raj *Jul 2014 – May 2015*

- **Screened natural compounds for activity:** Analyzed leaf essential oils from *Cinnamomum tamala* and *Cinnamomum verum*, ran molecular docking, and used ex vivo superfusion assays on rat ileum to show the spasmolytic activity of eugenol.

Patent

Chakraborty, A., Wallace, D.P., and Yu, A.S.L. "CRISPRa Compositions, Methods of Use Thereof for Treating Polycystic Kidney Disease, and Methods of Making the Same." U.S. Provisional Patent Application No. 64/034,047, filed April 9, 2026 (patent pending).

Publications and Preprints

Peer-Reviewed Journal Articles:

- **Chakraborty, A.**, Varghese, M.M., Wallace, D.P., Ward, C.J., Yu, A.S.L., 2026. CRISPR activation of endogenous PKD1 increases polycystin-1 levels and suppresses cellular features of ADPKD. American Journal of Physiology-Renal Physiology, 331, pp.F685-F699. https://doi.org/10.1152/ajprenal.90084.2026 *APSselect Award, November 2026.*
- **Chakraborty, A.** and Yu, A.S.L., 2026. Miniaturization of CRISPRa plasmids for efficient delivery into renal epithelial cells and Pkd1 transactivation. Molecular Biology Reports, 53(707). https://doi.org/10.1007/s11033-026-11872-1
- Thornton, M., Sommer, N., McGonigle, M., Kumar Ram, A., Yerrathota, S., Ehirim, H., Chaturvedi, A., Dinh Phan, J., **Chakraborty, A.**, ...Sharma, M., 2025. Notch3 deletion regulates HIV-1 gene expression and systemic inflammation to ameliorate chronic kidney disease. Disease Models & Mechanisms, 18(2), p.DMM052056. https://doi.org/10.1242/dmm.052056
- **Chakraborty, A.** and Yu, A.S.L., 2025. Prospects for gene therapy in polycystic kidney disease. Current Opinion in Nephrology and Hypertension, 34(1), pp.121-127. https://doi.org/10.1097/mnh.0000000000001030
- **Chakraborty, A.**, Sankaran, V., Murugan, R. and Chellappan, D.R., 2021. Comparative spasmolytic effect between Cinnamomum tamala and Cinnamomum verum leaf essential oils and eugenol through in vitro and in silico approaches. Zeitschrift für Naturforschung C, 76(9-10), pp.383-391. https://doi.org/10.1515/znc-2020-0153
- **Chakraborty, A.** and Viswanathan, P., 2018. Methylation-demethylation Dynamics: Implications of Changes in Acute Kidney Injury. Analytical Cellular Pathology, 2018(1), p.8764384. https://doi.org/10.1155/2018/8764384
- **Chakraborty, A.**, Sankaran, V., Ramar, M. and Chellappan, D.R., 2015. Chemical analysis of leaf essential oil of Cinnamomum verum from Palni hills, Tamil Nadu. Tic, 3(e10), p.x10.
- Sankaran, V., **Chakraborty, A.**, Jeyaprakash, K., Ramar, M. and Chellappan, D.R., 2015. Chemical analysis of leaf essential oil of Cinnamomum tamala from Arunachal Pradesh, India. Journal of Chemical and Pharmaceutical Sciences, 8(2), pp.246-248.

Accepted Manuscripts and Preprints:

- Nyimanu, D., **Chakraborty, A.**, Parnell, S., Wallace, D. and Yu, A., 2026. Apelin inhibits cyst growth and improves kidney function in mice with polycystic kidney disease. bioRxiv. Accepted in Kidney360 (in press). https://doi.org/10.64898/2026.03.26.714294

Fellowships, Honors & Awards

- **APSselect Award:** American Physiological Society, for first-author paper in American Journal of Physiology-Renal Physiology (Nov 2026).
- **American Heart Association (AHA) Predoctoral Fellowship:** Awarded $71,828 national fellowship grant (Jan 2024 – Dec 2025). https://doi.org/10.58275/AHA.24PRE1194472.pc.gr.190750
- **ASN Kidney STARS Travel Award:** American Society of Nephrology (ASN) Kidney Week, Houston, TX (2025) & Philadelphia, PA (2023).
- **ASN Kidney TREKS Scholar:** Selected for MDBIL Research Program, Bar Harbor, ME (2024).
- **Biomedical Research Training Program Scholarship:** Awarded $13,000 – University of Kansas Medical Center (2023).
- **KUMC Physiology Society Travel Award:** University of Kansas Medical Center (2023).
- **Best Poster Award:** 4th Annual Internal Medicine Research Day, University of Kansas Medical Center (2023).
- **Second-Best Poster Winner:** Student Research Forum, University of Kansas Medical Center (2023).
- **Student Governing Council Travel Award:** ASN Kidney Week, Orlando, FL (2022).
- **Elsevier Best Research Paper Award:** 43rd Annual Conference of Environmental Mutagen Society of India (2019).

- **CSIR Senior Research Fellowship:** Council of Scientific and Industrial Research, Government of India — nationally competitive research fellowship held at Vellore Institute of Technology (Dec 2017 – Jun 2019).
- **Gold Medalist:** Best Research Paper, SRM University Research Day (2017).
- **Student Athlete:** Selected to represent the School Games Federation of India (SGFI) at 4 National Field Hockey Championship Tournaments (2006 to 2010).

Scientific Presentations

Invited Oral Presentations:

- **Chakraborty, A.** and Yu, A.S.L. Miniaturized CRISPR activation plasmid improves transgene delivery, PKD1 expression and induces signaling correction in polycystic kidney cells. Jared J. Grantham Symposium, Kansas City, KS (2025).
- **Chakraborty, A.** and Yu, A.S.L. Epigenetic reprogramming of PKD1 transcription: A proof of concept. Jared J. Grantham Symposium, Kansas City, KS (2023).
- **Chakraborty, A.** and Viswanathan, P. Ischemic acute kidney injury is characterized by heterogenous methylomic changes in the renal cortex and medulla. 43rd Annual Conference for Environmental Mutagen Society of India, Vellore, TN, India (2019).
- **Chakraborty, A.**, Devaraju, P., Jayaraman, M. Analysis of selected upstream variants in TNF-alpha gene for their association with gastric mucosal inflammation in Indian Tamil patients with chronic Helicobacter pylori infection. SRM University Research Day, Chennai, TN, India (2017).

Poster Presentations:

- **Chakraborty, A.**, Wallace, D.P., Yu, A.S.L. Rationally engineered miniaturized plasmids improve transgene delivery and CRISPR-activation of PKD1 in primary kidney cells. ASN Kidney Week, Houston, TX (2025).
- **Chakraborty, A.** and Yu, A.S.L. Rationally Engineered Miniaturized Plasmids Improve Transgene Delivery and Expression in Hard-to-Transfect Primary Kidney Cells. Student Research Forum, KUMC, Kansas City, KS (2025).
- **Chakraborty, A.**, Ward, C.J., Yu, A.S.L. CRISPR activation of PKD1 in immortalized cell lines is limited by its heterochromatinized proximal promoter. ASN Kidney Week, Philadelphia, PA (2023).
- **Chakraborty, A.**, Ward, C.J., Yu, A.S.L. Heterochromatinized proximal promoter of PKD1 limits CRISPR activation of the gene in immortalized cell lines. PKD Connect Conference, Denver, CO (2023).
- **Chakraborty, A.**, Parnell, S.C., Zhang, Y., Daniel, E., Vrtek, K., Raman, A., Reif, G., Wallace, D.P. Early cyst formation leads to the development of an inflammatory microenvironment and tissue remodeling in polycystic kidney disease. ASN Kidney Week, Orlando, FL (2022).
- **Chakraborty, A.**, Ramar, M., Chellappan, D.R. Elucidation of receptor mediated action of eugenol from Cinnamomum tamala and Cinnamomum verum leaves against ileal spasms. 41st Indian Society of Human Genetics Annual Meeting, Chennai, TN, India (2016).
- **Chakraborty, A.** and Ramaswamy, R.M. Ex-vivo Characterization of Atheromatous Tissue Under SPION Based Magnetic Hyperthermia. International Conference on Emerging trends in Bioengineering, SASTRA Deemed University, TN, India (2013).

Academic Leadership

- **Planning Committee Member** | Polycystic Kidney Disease-Renal Resource Consortium (PKD-RRC) Symposium (Apr 2023)
  - Co-managed speaker scheduling and technical execution for a national level scientific symposium, coordinating cross-institutional participation for renal researchers and trainees.
- **Vice President** | Cell Biology & Physiology Society, University of Kansas Medical Center (Dec 2021 – Dec 2022)
  - Led executive operations for departmental graduate trainees, organizing scientific seminars, professional development workshops, and networking events to enhance student career preparation and inter-departmental collaboration.

- **Interprofessional Committee Member** | Student Governing Council, University of Kansas Medical Center (Aug 2022 – Aug 2024)
  - Represented graduate student interests across medical and health professions schools, evaluating inter-professional grant proposals and advocating for campus-wide research resources.
- **Organizing Committee Member** | 3-Minute Thesis, Student Research Forum, University of Kansas Medical Center (Jan 2022 – Dec 2024)
  - Directed annual university-wide science communication competitions over a 3-year term, managing judge recruitment, workshop training for participants, and event execution to improve graduate student communication skills.
- **Chair** | 8th Genetic Engineering Research Symposium, SRM Institute of Science and Technology (Apr 2016)
  - Directed a multi-committee team to execute a national research symposium, managing budget allocation, abstract peer-review, and keynote speaker invitations for 100+ attendees.

Certificate and Professional Memberships

- **Professional Membership:** American Society of Nephrology (ASN) – Active Member (2018 – Present).
- **Professional Membership:** American Association for the Advancement of Science (AAAS; Mar 2024 – Feb 2025) & American Heart Association (AHA; Jan 2024 – Jun 2025).
- **Coursera / Johns Hopkins University:** Introduction to Genomic Technologies (2020). https://www.coursera.org/account/accomplishments/verify/66YYT3UT4KTQ
- **Coursera / University of Melbourne:** Epigenetic Control of Gene Expression (2018). https://www.coursera.org/account/accomplishments/records/DX9LW662XBUX